

EDUCATION

- **MSc Machine Learning and Data Mining** Aug. 2023 - Sep. 2025
University of Jean Monnet, Saint-Étienne, France
- **MSc Computer Science (ERASMUS+ Semester)** Oct. 2024 - Mar. 2025
Albert-Ludwigs-Universität Freiburg, Germany
- **B.E. Computer Science and Engineering** Aug. 2017 - May. 2021
Sri Venkateswara College of Engineering, Chennai, India

EXPERIENCE

- **AI Researcher, Intern** Marseille, France
Institute of Developmental Biology of Marseille Apr. 2025 - Aug. 2025
 - **Research:** Worked on interpretable large language models (LLMs) to classify cellular states from bulk and single-cell RNA-seq data. Integrated explainability frameworks to identify key genes influencing cell classification, improving classification accuracy and biological interpretability by over 5% compared to baseline models.
- **Student Researcher, Part-Time** Freiburg, Germany
Bioinformatics group, Albert-Ludwigs-Universität Freiburg Nov. 2024 - Mar. 2025
 - **Research:** Built BERT- and GRU-based deep learning pipelines to predict neoantigen immunogenicity on experimentally validated datasets, achieving 80% classification accuracy.
- **ML Researcher, Intern** Saint-Etienne, France
Laboratoire Hubert Curien Apr. 2024 - Jul. 2024
 - **Research:** Enhanced graph learning models using screening-based sparsity methods, reducing runtime by upto 20%. Contributed to research on interpretable structured learning under varying optimization constraints.
- **Programmer Analyst** Chennai, India
Cognizant Technology Solutions Jul. 2021 - Aug. 2023
 - **Engineering:** Engineered and deployed full-stack web applications using Angular, JSP, and SQL Server for enterprise clients. Collaborated across four agile teams as a DevOps developer managing databases including PostgreSQL and MongoDB, and maintaining build pipelines.

PROJECTS

- **Post-hoc Explainable AI (XAI) for Protein Subcellular Localization (In Progress):** Fine-tuned the pretrained *cytoself* model on human protein images. Currently applying post-hoc XAI techniques including latent-space visualization and feature attribution methods to interpret model predictions.
- **Neoantigen Immunogenicity Prediction:** Improved classification accuracy by training a BERT-based architecture on 20k peptide sequence samples, outperforming a GRU-based classifier.
- **Structured Generation from Call Transcripts using FastAPI and LLaMA:** Developed a web application to generate call transcripts and extract structured information. Built backend APIs using FastAPI and integrated LLaMA / OpenAI APIs with prompt engineering. Designed an interactive frontend using HTML, CSS, and JavaScript.
- **Transfer Learning using ResNet:** Fine-tuned a pretrained ResNet model on the MNIST dataset for digit classification using transfer learning techniques.
- **Library Management System with Book Recommender:** Developed an end-to-end application featuring a collaborative filtering-based book recommendation system. Implemented backend services using Flask and designed interactive frontend interfaces with JavaScript.

SKILLS

- **Programming:** Python, Java, C++, SQL
- **Machine Learning & AI:** Deep Learning, Machine Learning, NLP, Large Language Models (LLMs), Computer Vision, Retrieval-Augmented Generation (RAG), Explainable AI (XAI)
- **Frameworks & Libraries:** PyTorch, TensorFlow, scikit-learn
- **Tools & Platforms:** Git, Docker, FastAPI, HPC, Maven, Udeploy, Apache Subversion